Technical report — Gegenprobe: one schema and a converter for Swiss parliamentary answers

Track 2A — OpenParlData · Online · Team 444 — Roman (+ Claude Code agents, §7) · Evidence site https://gegenprobe.mi11ion.io

maturity	components
ready to apply · ready to deploy	census and Bern repair lists (relinks: one OPD decision, its hash rule) · nightly job (three real nights on standard GitHub runners: one with the venv unit, 10 Oct, and two consecutive with the Docker unit, 11 Oct)
audited · tested prototype	source-adapter lists · rules-only converter
feasibility study · drafts for the eCH group	review mode, whole and virtual affairs · schema v0.1 and v0.2 (P8's roles)

1. Summary

Can one schema hold Swiss parliamentary business? Yes, if each field says which role its value plays (e.g. "answer date" = the government's decision, its letter or the submission date). Asked of interventions, OPD's five intervention types: 198,594 of its 325,691 affairs; parliamentary initiatives (5,817) and government business (37,864) are outside the tested scope. *Shared:* one schema of 17 fields for communal, cantonal and federal business; every value carries its role and source (page and span, or the OPD record), every gap a typed reason. *Needs roles:* **10 fields**, by source of the split 4 parliamentary practice, 4 partly OPD's tagging (which OPD can correct), 2 open; 2 are fragile (the split holds under one of P8's two checks). Post hoc, in P10's gold from 15 parliaments none of the 197 people needed the guide's open `other` role (0/197). *Body-specific:* **1 field and 2 constructs**: the submission date (Vernier and Lausanne print theirs nowhere); oral answers, councils that publish only minutes. *Untested:* **6 fields**, defined once in v0.1; meaning is measured for the other 11: 4 on OPD's labels, 6 on the documents of 66 affairs from 11 parliaments (P8), 1 (type) on both.

Delivered: repairs for 9,720 answers keyed on OPD ids, every one from model-free rules; schema v0.2 with the roles; a converter that gives every value its page and span; a nightly job. Apertus only picks values printed in the document, and code locates them (70B weights via the Public AI gateway; local 8B GGUF on CPU). **Runtime external services:** none by default (public sources only); Apertus review mode sends document text to the Public AI gateway, or nothing with the local 8B.

92.3%	of interventions with a recorded government answer would have answer text once the proposed repairs are applied, up from 81.6% today: 91.6% from row-checked repairs + 0.7 points from the 592 agenda joins and 39 review rows
9,720	answers with a model-free repair proposed to OPD, keyed on OPD ids; the Bern plan ran on PostgreSQL: 2,975 rows in one transaction that rolls back on any deviation
10 of 17	fields are shared once their roles are explicit; 1 is body-specific; 6 are not yet tested for meaning
24/29	answer dates Apertus picks right among the printed dates in 15 parliaments nobody wrote rules for: the rules 2/29 (registered), a generic keyword rule 16/29 (post hoc)
50/141	author roles Apertus reads right in 14 parliaments without rules (rules 17/141); the gain is the model-free candidate finder: document order 52/141 (post hoc), level
120	records the converter made, from 23 parliaments (communal, cantonal, federal), validate in schema v0.2; so do P8's 66 gold records

Sources: 92.3% and 9,720 from the census and the repair lists (no model); 10 of 17 from OPD's labels and P8 (pre-registered); 24/29 from P2, registered (the keyword rule post hoc, proposed in review: exact McNemar p = 0.057); 50/141 from P10, registered (recall +23.4 points over the rules, Holm p = 0.032), level with OPD's contributor list (37/141; H3a, Holm p = 0.17); 120 from P3, P3b, P5 and the federal cell, whose 15 affairs are exploratory.

Glossary. **P1–P8, P10**: the ten tests (P3b replicates P3), each frozen by hash before its labels (§6). **D-numbers**: logged deviations from a registration. **θ**: the frozen margin above which Apertus answers null. **Branch B**: P1's registered outcome that keeps Apertus off the unattended path. **Post hoc**: analysed after the labels, often proposed in review; **registered**: decided by the frozen rule.

1. **Schema.** P8 finds roles in authorship, co-signatory lists, the instrument behind a harmonised type and the language (text, OPD's tag and the affair's can differ). The converter fills v0.2's roles with fixed rules (75 of its 105 records carry printed authors with their role); on 5 parliaments its rules never saw, the language is right on 60/60 documents and author (name, role) pairs reach precision 39/49 at recall 39/137 (exploratory). Tables, annexes, images, scanned pages and signatures are marked by page range (66 of 105 converter records carry one); in a blind check of 30 documents (108 pages; exploratory, in-sample for the detector) tables are found at page precision 12/12 and recall 12/15, annexes at 6/6 and 6/6.

2. **Ready for OPD.** Bern's 2,421 ready relinks and 48 ready fetches, from a linking bug OPD confirmed (correspondence, 10 Oct 2026); 554 `begin_date` corrections; lists from 50 model-free source adapters (blind AI annotators: 197/200 are the executive's answer); 9 OPD vocabulary labels wrong as published; a Fedlex resolver for 29,946 federal placeholder documents (§5).
3. **Apertus** earns its place choosing the right value among printed candidates. *The answer date:* without rules it finds **24/29** answer dates, the rules 2/29 (P2, registered; by parliament the gain is [-0.02, 0.54]), a generic keyword rule 16/29 (post hoc); on the review queue 95.8% of its promotions are answers whose printed answer date passes the date check, 73.2% for a no-model exact-date match (P7-H4, registered, weighted). *Who is an author* (P10, registered: recall +23.4 points over the rules, precision non-inferior; not significantly better than OPD's list, H3a): post hoc, the gain over the rules comes from the model-free candidate finder plus document order (52/141, F1 0.41; Apertus 50/141, F1 0.42), and Apertus adds precision deciding who is an author (+7.3 points [0.8, 18.2], uncorrected). Where rules exist it adds about one document (P1: cascade 139/154, model-free cascade 138/154, post hoc).

2. Architecture

Census (`census/`) streams the OPD exports once into a per-affair *answer-link status*, the feasibility matrix and a comparability measure; a derived layer (`census/corrected.py`) places every registered gap row once the repair routes have looked (Fig. 1). **Repair** (`repair/`): deterministic routes per cause, each row checked (affair number or title, answer heading, date against OPD's answer event), as lists keyed on OPD ids; a nightly job (`deploy/nightly/`) re-runs the Bern route. **Converter** (`convert/`, `qa/`): OPD metadata and events, then frozen DE/FR/IT anchor rules, then the Apertus closed-set pointer on the residue; modes `rules_only` (default, no model) and `review` (suggestions with page, span and margin, flagged `needs_review`); `python -m gegenprobe affair --opd-affair <id>` turns one OPD affair into one validated v0.2 record, with OPD's text spans (`opd_text_span`) and Q/A pairs. **Schema and validator** (`schema/`, `validate/`): Pydantic schema v0.1, normative; a value carries its role, page, span, quote hash, route and rule, a null its typed reason (`data/schema/null_reasons.json`); `QAItem` pairs question *n* with its answer passage.

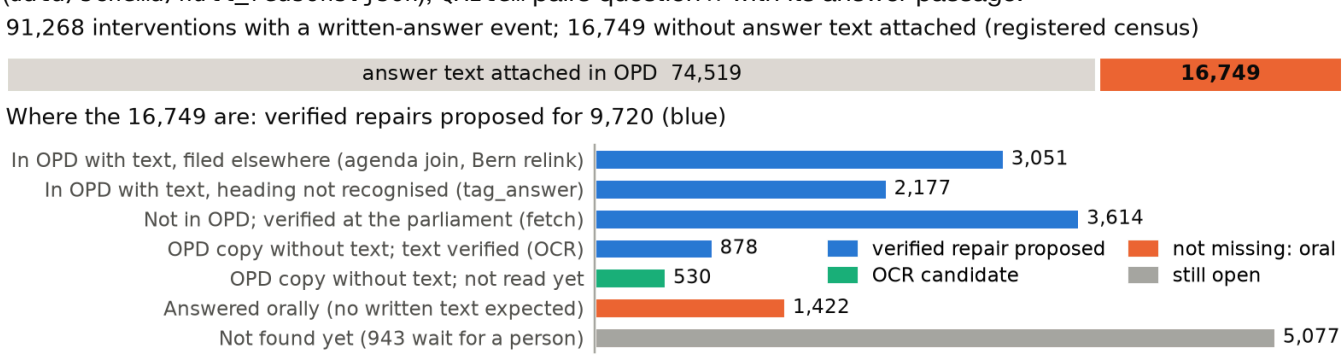


Figure 1. The registered 16,749 placed once each (`results/census/corrected.json`). Still to find: 5,077 (943 of them already waiting for a person).

3. Use of Apertus

Closed form only. Apertus never writes a value. (i) *Verifier* (P7, `repair/adapters/adjudicate.py`): for an adapter review row it answers "does this document print the executive's answer to this intervention (number, title)?" (yes vs no scored after a JSON prefix, θ_A = 8) and picks the answer date among the printed dates or none (θ_B = 10); a row is promoted only if both pass and the date passes the adapters' own date rule. (ii) *Pointer* (`convert/pointer.py`, prompts `src/prompts/{de,fr,it}`): the document (first 12,000 characters) and the question in the user message, the assistant turn prefilled with a JSON prefix, `null` and every distinct printed date scored by summed log-probability; null if `lp(null) − logsumexp(lp(dates)) > θ`. Code locates every span (`pointer.locate`). θ does not transfer across builds, so it is fitted per build on dev documents and frozen (θ = 10.40, AUROC 0.99). **Why closed:** asked freely, Apertus dates 15/15 (8B) and 14/15 (70B) Bern interventions that have no answer.

Which model. Evaluation used **Apertus v1.5 70B weights served by Public AI** under the name `swiss-ai/apertus-v1.5-8b` (inferred from the served fingerprint `tp8-pp2-…` and 760/792 greedy outputs byte-identical to the `-70b` name); the response `model` is checked on every call. The **8B** (`swiss-ai/Apertus-v1.5-8B`, a community 4-bit GGUF, sha256-pinned) runs locally on CPU with its own θ = 16.73 (exploratory): on P2's unseen parliaments the registered P2-H2 test over the rules gives 13 vs 0; its two-stage reader v2 cuts dates on answerless documents from 18/34 to 5/34 (P6). **Recommendation:** `rules_only` unattended; for suggestions the 8B where no text may leave the machine, the 70B for accuracy on infrastructure the operator chooses. Review mode stops (exit 3) on a served build θ was not fitted on.

Cost (`results/report/cost_end_to_end.json`, one core, no GPU). Fetch, extraction and checks about 2.1 s per document plus 6.9 s per OCR page; the rules process 480 documents per second (0.5 CPU-hours for all 902,754 documents in OPD's export; at ten times OPD's volume about 5 CPU-hours). Review mode: USD 0.59 per 1,000 residue documents on the 70B; 180 s per document on 4 vCPUs on the 8B, so the estimated 11.5 residue documents a night (18.2 answers a day in 2025, residue share 0.63) take about 34 minutes.

4. Data and operations

Data. OPD bulk exports (snapshot 6 Oct 2026; CC BY 4.0, "Source: OpenParlData.ch"), Bern open data, Fedlex, parliament sites, fetched politely (≥ 1.2 s per host and process). `data/` and `results/` are CDLA-Permissive-2.0 and carry OPD's attribution. **Sovereignty and data protection.** No third-party processor by default; besides the public sources, Hugging Face once for the pinned 8B, and a review-model endpoint (Public AI, or the local 8B, where no text leaves the machine) and an alert relay only if the operator sets them. The operator sheet's nDSG row names the personal data (office holders named in public documents), the route that discloses text abroad (Public AI) and the cache retention (default 30 days); for public bodies it recommends rules only or the local 8B. **Operations.** `make run` and the nightly unit run the image read-only, as a non-root uid, with `--cap-drop ALL`; OCR fails loudly (a typed `ocr_failed`, never an empty page); with the pinned deu/fra/ita models all 154/154 P1 documents re-derive from the source PDFs, the 22 Ticino scans included. Nightly job (three real network nights, top table): the ledger-status bug night 2 exposed is fixed and covered by a two-night test. A network night refuses to start without a contact URL in `GP_USER_AGENT`. **robots.txt:** we found, audited and fixed our own robots.txt bug; every fetch path now fails closed (disclosure: `docs/incidents.md`).

5. Results without a model: schema verdict, census, repair

Table 1. The 17 fields. *Class*: role split = one label covers several roles, measured on OPD's labels (`results/census/comparability.json`) or on the documents (P8, `results/p8/`); fragile = not under both P8 checks; shared = one v0.1 definition. *Source* of the split (committed results only): practice (parliaments differ; the standard encodes the role), OPD (likely a data or tagging error OPD can correct), both, or open (the labels cannot tell). *Where it lives*: parliaments where OPD metadata carries it · carries it for part of the affairs · only the PDF · not in OPD's data (`results/census/matrix.json`).

field	class	source	evidence	where it lives (meta · mixed · PDF · none)
answer date	role split	both	decided, dispatched or registered; in 7 of 54 bodies the submission date	**metadata**: 53 · 1 · 0 · 0
procedural history	role split	open	35 event labels, several date roles	**metadata**: 78 · 9 · 1 · 1
type	role split	both	42 of 1,038 raw labels map to two types; P8: the instrument	partly: 61 · 24 · 2 · 2
procedural state	role split	open	32 labels ambiguous	partly: 41 · 27 · 18 · 3
decision outcome	role split	practice	3 labels ambiguous	partly: 7 · 31 · 33 · 6
submitter	role split	practice; OPD in GR	P8: sole, joint or group author	partly: 23 · 61 · 5 · 0
co-signatories	role split	practice	P8: includes authors of record	PDF only: 0 · 2 · 72 · 12
language	role split	both	P8: tag names the other language; text, tag and affair differ	partly: 14 · 53 · 20 · 2
addressed body	split, fragile	practice	P8: the council (Bellinzona)	partly: 12 · 10 · 54 · 10
government recommendation	split, fragile	both	P8: by stage	partly: 3 · 8 · 56 · 10
submission date	body-specific in 2	OPD	P8: Vernier, Lausanne	**metadata**: 89 · 0 · 0 · 0
answer-link status	shared	—	defined once in v0.1	**metadata**: 54 · 0 · 0 · 0
decision date	shared	—	defined once in v0.1	partly: 7 · 30 · 34 · 6
Q/A pairing	shared	—	defined once in v0.1	partly: 22 · 25 · 4 · 3
usable text	shared	—	defined once in v0.1	partly: 41 · 42 · 0 · 6
dispatch date	shared	—	defined once in v0.1	PDF only: 0 · 0 · 47 · 7
related affairs	shared	—	defined once in v0.1	PDF only: 0 · 0 · 76 · 13

4 fields are in OPD's metadata almost everywhere, 13 partly or only in the PDF. Two further *concepts* are body-specific: oral answers in question time (23.6% of question-time answers) and councils that publish only meeting documents (16 bodies; v0.1 marks affairs rebuilt from minutes `virtual`, P4).

Cross-parliamentary question. Of the 91,268 interventions with a recorded government answer, 81.6% have answer text attached in OPD (the registered count); 82.3% also count the 592 answers OPD files under an

agenda item of the affair (not sample-checked); 85.6% with the repairs from what OPD already holds, 88.3% after the Bern relinks, 92.3% with the checked fetches (`results/report/r9_numbers.json`).

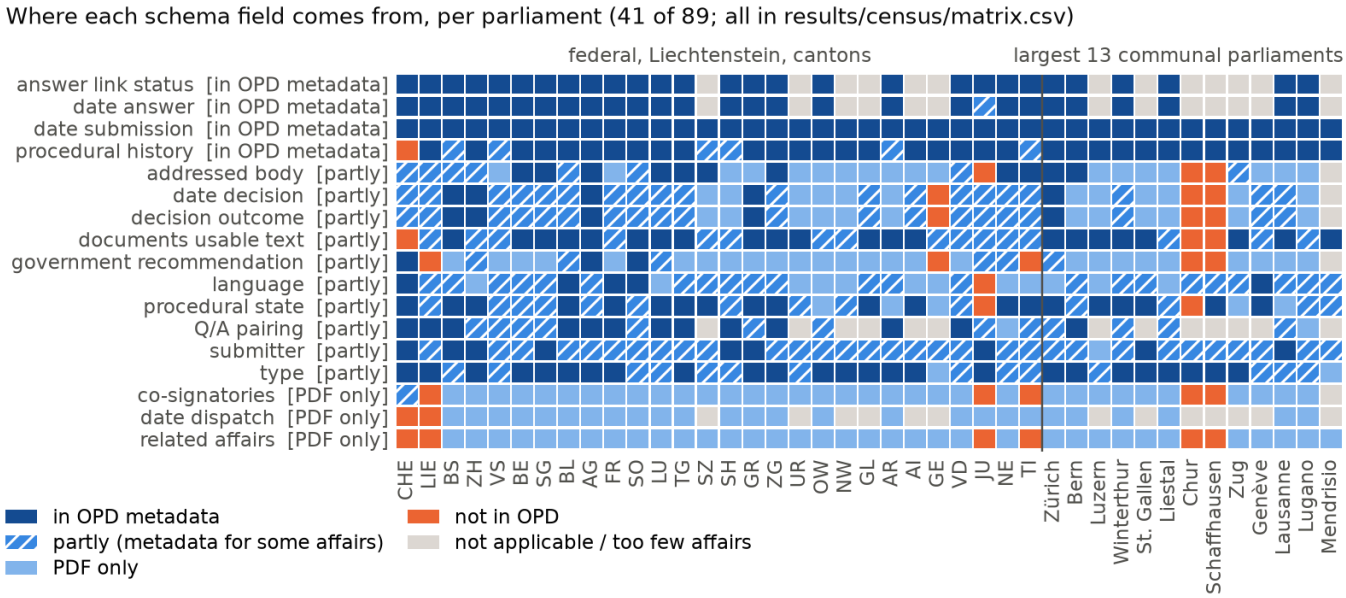


Figure 2. Where each field comes from, per parliament. Orange = not in OPD's data for that parliament (the parliament may still publish it).

Repair. *Bern*: of 3,532 answers the canton publishes OPD attaches 229 (6.5%), although 96.1% are in OPD and linkable without a model. 2,459 relink candidates = 2,421 that pass every check against OPD's own copy (affair number, heading, decision date = OPD's answer event [97.9, 98.9]) + 38 to review. The document `hash` is sha256(body_key + parent_type + url) (49,810 of 49,810 rows), so relinking changes each identifier and file URL unless OPD keeps the old one; the plan carries both values. *Source adapters* (`docs/adapters.md`): readers for 50 parliaments (27 yield counted rows) verify 6,613 answers by affair number or title, the parliament's answer formula and a date check. 2,177 of them are already in OPD with text under a heading the census lexicon does not list: the repair is a category on the document (`tag_answer`). Listed separately: 282 Jura answers matched by identity and wording only, and 917 review rows (214 of them recovered by Apertus into the tier `promoted_by_apertus`, §6). **Blind audit v3** (`results/repair/adapters/audit3/`; 200 counted rows, 60 decoys, key hashed on GitHub first): Claude Opus judged 197/200 the executive's answer (Wilson 95.7% to 99.5%; date wrong or unsupported on 15) and rejected 60% of the failed decoys; Qwen-SEA-LION accepted 200/200 (252 of 260 packets agree).

6. Evaluation: ten pre-registrations

Design. Each test froze code, prompts and θ by sha256 before any label was committed (P1's hypothesis script followed its labels, D4). Unit (P1, P2, P6): one document's answer date, strict. Arms on identical inputs: **rules**; **pointer**; **cascade** (rules, then the pointer on the residue); **naive** free generation, plain and null-allowed; trivial baselines. **Gold, no human labelling by design:** two blind agent annotators (Claude Opus, Sonnet), a third breaks ties, quotes verified by code, raw files committed; all 96 annotator sessions (P1–P8, P10, the document-parts check, the federal cell) ended before GitHub received their labels (`results/eval/annotation_sessions.json`).

Table 2. The ten tests, in registration order.

test	n	what it tests	registered result	note
P1	154 documents, parliaments with rules	answer date: rules vs Apertus	**partial**: H3 held; H1 null safety failed (7/90); H2 Branch B	rules alone 135/154, so no model by default
P2	62 documents, 15 parliaments without rules	rules transfer; pointer vs rules	**3 of 4 held**; H3 null safety failed (4/33)	pointer 24/29 dates, rules 2/29
P3	29 whole affairs, 10 parliaments	7 fields: metadata, rules, Apertus	**partial**: no field unattended; Apertus recall held, precision below	feasibility study (D3-5)
P3b	36 fresh affairs, 9 parliaments	replication of P3	**held**: replicated (S1, S2)	S2 lenient; type 35/36; addressed body 2/33
P4	120 agenda items, 6 sittings	affairs rebuilt from minutes	**partial**: H2, H4 held; H1, H3 failed	recall 99/120, precision 99/99
P5	40 fresh affairs, 10 parliaments	Q/A pairs, people, recommendation	**partial** tier (H1); H2, H5 held; H4 as rules vs free generation; H3 no gain	*not yet* when parliaments are resampled
P6	68 fresh documents, 8 parliaments	answer-date reader v2 vs free prompting	**no hypothesis held**	free prompt 62/68, v2 59/68

test	n	what it tests	registered result	note
P7	478 review rows, 190 blind packets	Apertus as verifier of the review queue	**H1 holds** (120/120); H4 supported; H3 not supported	214 of 917 rows promoted
P8	66 fresh affairs, 11 parliaments	meaning of 7 fields in the documents	**none uniform**: 6 role split, 1 body-specific	5 robust; meaning measured for 11 of 17
P10	34 fresh affairs, 14 parliaments without rules	author roles: Apertus reader vs rules, OPD's list	**3 of 4 held**: H1, H2, H3b; H3a not	pairs right 50/141, rules 17/141, OPD 37/141

Unattended (95% lower bound ≥ 0.80, document-level intervals): the rules' answer date where rules exist (P1, helped by right nulls: dates found 47/64), the type on whole affairs (P3b, 35/36), and in P5 question detection (recall 167/187, precision 167/175), the reasoning span (37/40) and co-signatory precision (101/103). **Cross-checks.** `Qwen-SEA-LION-v4-32B-IT` (AI Singapore) relabelled P1, P2, P6 and P7 blind: date κ 0.80, 0.87 and 0.86, P7's is-answer labels 185/190; no registered decision changes. Non-LLM checks: OPD's event matches 37 of 44 checkable P1 dates; P6: 10/10, every gold quote on its page (34/34). Holm over all 20 registered confirmatory tests of P1–P7: the 9 that held survive (`results/eval/posthoc_r8_multiplicity.json`); whole parliaments are resampled as well.

P1 arm (154 documents)	exact [95% CI]	answered: found /64	wrong printed date	invented	answerless dated /90
rules only (default)	**135/154 [81.5, 92]**	47	4	0	2
pointer	130/154 [77.9, 89.3]	46	16	0	6
cascade	139/154 [84.6, 94]	56	11	0	7
naive, null-allowed	134/154 [80.8, 91.4]	47	11	0	3
trivial: latest date, answer-typed docs	*131/154 [78.6, 89.8]*	*44*	*16*	*0*	*3*
model-free cascade (post hoc)	*138/154 [83.8, 93.5]*	*52*	*11*	*0*	*4*

P1: the rules (135) and the latest date on answer-typed documents (131) are level (paired p = 0.48, post hoc; P1's rules entered with its sample, D12), so P2 is the test of transfer. H1 null safety failed (the cascade dates 7 of 90 answerless documents, 6 of them council reports; upper bound 15.2% vs 10%); H2 gave Branch B (cascade alone right on 5, rules alone on 1, p = 0.22), so Apertus left the unattended path for an opt-in review suggestion (D6); H3 held (AUROC 0.969).

P2 arm (62 documents)	exact [CI]	answered: found /29 [CI]	wrong date	invented	answerless dated /33
rules only	35/62 [44.1, 68.1]	**2/29** [1.9, 22]	2	0	0
pointer	**53/62** [74.7, 92.2]	**24/29** [65.5, 92.4]	8	0	4
naive, null-allowed, re-parsed (D2-4)	54/62	24/29	7	1	3
trivial: latest printed date	*24/62*	*22/29*	*38*	*0*	*31*
trivial: latest date, answer-typed docs	*44/62*	*18/29*	*13*	*0*	*7*

P2: the rules collapse off their home layouts (47/64 → 2/29). H2 held: on all 62 documents the pointer beats the rules on 22 and loses on 4 ($p = 5.3 \times 10^{-4}$); resampling the 15 parliaments, the gain is [-0.02, 0.54], so it sits in a few parliaments. "Answerless" means no qualifying answer date: the pointer's 4 such dates fall on documents with an answer but no qualifying date; of the 25 with no answer at all it dates none (post hoc; with P6's v2 none of 53 across P2 and P6; P6's v1 dated 1 of 28). *Post hoc, proposed in review:* a generic DE/FR/IT keyword rule (answer cue, first date after a sitting cue, else the latest; answer-typed documents) is exact on 45/62 and finds 16/29 answer dates (pointer vs rule: McNemar p = 0.077; on answer dates 0.057).

P3/P3b/P4. Default (metadata + rules) on 29 affairs: type 27/29, answer date 22/27, 27/27 with Apertus suggestions. P3b replicated by its registered criteria; answer date 20/35 by default (C), 31/35 with Apertus suggestions. P4: 99 of 120 agenda items as virtual affairs, precision 99/99, no decision invented (0/36). **P5.** Rules-only pairing: precision 118/143 = 0.83, recall 118/176 = 0.67, *partial* with document-level intervals, *not yet* when parliaments are resampled (precision [0.66, 0.97]); Recommendation 34/40 by rules; submitter rules 12/40, pointer 24/40, free generation 34/40.

P6 arm (68 documents, 34 answered)	exact	answered: found	answerless dated	printed nowhere
free prompt, null allowed	**62/68**	32/34	4/34	0
cascade v2 (rules, then v2)	60/68	31/34	5/34	0
v2	59/68	30/34	5/34	0
rules only	54/68	20/34	0/34	0
trivial: latest date, answer-typed docs	*51/68*	*25/34*	*8/34*	*0*

P6 — reader v2 (an answer-present gate, then the gold guide's decision-before-letter-date rule). No registered hypothesis held: v2 vs free prompting 1:4 (p = 0.38); v2 dates 5/34 answerless documents (upper bound 30.1% vs 15%), all 5 containing an answer without a qualifying date; v2 vs v1 5:2 (p = 0.45).

P7 — Apertus as a precision verifier of the review queue (`eval/prereg8/`, LOCK8 before the draw). Test frame: 478 eligible review rows outside every audit; 190 blind packets, gold A1 Opus / A2 Sonnet. H1 (primary): promotion precision 120/120, Wilson lower bound 96.9% ≥ 0.90: **holds**. H4: dated precision 95.8% vs 73.2% for a trivial exact-date match, +22.7 points [10.2, 34.3], weighted, as registered (unweighted +7.5): **supported**. H3: the rows free generation with a printed-date filter adds are 28/30 right: **not supported**. Base rate: 86.7% of eligible queue rows are answers, but only 40.9% are answers whose own date passes the rule, so the gain is the date check. *Post hoc* (`results/p7/`): the metric measures how often promotions are date-checkable, not the date picked. A model-free rule accepting any printed date of the same closed set reaches 76.2%: P7 +19.6 points [8.4, 30.9] (parliament-cluster [0.9, 42.8]); Apertus's chosen date is the gold date on 113/120 of its promotions, the exact-date match's on 127/146 (its promotions are answers on 146/146), each on its own promotions (not paired). The rule recovers more rows with less precision (attainable recall 88% vs 66%; 369 vs 214 queue rows). Applied to the whole queue, 214 of 917 rows go to the tier `promoted_by_apertus`, a pre-sorted batch for review.

P8 — what the shared fields mean (`eval/prereg9/`, LOCK9 before the draw). Blind labels of what seven fields denote in the two documents OPD links. Share in v0.1's meaning, pooled (parliament-cluster 95% interval): submitter 35/65 [33.3%, 75.0%], co-signatories 10/41 [6.8%, 46.3%], submission date 44/57 [57.1%, 91.8%], type 55/64 [76.2%, 95.3%], language 34/36 [83.3%, 100.0%], each verdict unchanged on agreed labels only and on Qwen's labels; addressed body (57/60) and recommendation (31/34) are fragile.

P10 — Apertus reads author roles where no rules exist (`eval/prereg10/`, LOCK10 before the draw and the labels; D1). 34 fresh German and French affairs (of 38 planned; French 4 short under the registered fill rule), 14 parliaments outside every rule set, 141 gold (name, role) pairs (A1 Opus / A2 Sonnet, pair F1 0.868; A3 on 9 of 40 packets). Pairs right / printed: Apertus 50/96 (precision 0.52, recall 0.35, F1 0.42), rules 17/55 (0.31, 0.12, 0.17), OPD's contributor list as first author and co-signers 37/104 (0.36, 0.26, 0.30). Confirmed: H1, recall over the rules, +23.4 points [5.9, 49.0], Holm p = 0.032; H2, precision non-inferior (+0.21); H3b, over the same names with a trivial role. Not confirmed: H3a, over OPD's list (+0.12, Holm p = 0.17): where OPD lists the authors, its order is a strong baseline. *Post hoc, proposed in review:* the same candidates in document order with OPD's registered mapping match 52/141 (111 printed; 0.47, 0.37, F1 0.41). The gain over the rules is the candidate finder plus document order (H3b's registered baseline reads every candidate as a joint author: 16/141); Apertus's author filter, on the same order rule, keeps recall and adds +7.3 points precision [0.8, 18.2] (uncorrected). Shipped as an opt-in review route (`affair --roles apertus`): the reader alone; a fallback to OPD's list where it suggests no author is opt-in, because on P10's arms it added 32 pairs, none right (post hoc).

Federal cell (exploratory, `eval/fed/`, LOCKFED before draw and labels): 15 Federal Council answers (5 per language) print no answer date and no author; rules, pointer and converter date none (15/15 right nulls), the latest printed date dates an answerless document on 6/15; the instrument's function keeps one meaning (15/15); all 15 v0.2 records validate (with P3, P3b and P5: 120 converter records from 23 parliaments).

What the tests say. *Provenance by construction:* no arm that picks from printed dates returned a date printed nowhere (P1, P2, P6), close to Apertus asked freely with null allowed (P2 53/62 vs 54/62; P6 59/68 vs 62/68), with a page, span and margin on every value. **Null safety** is left to the reviewer: the registered null-safety tests on the 70B failed (P1-H1, P2-H3, P6-H2), which is why the default runs no model.

7. Next steps, reproducibility, declarations

Next steps. Q/A pairs are tested on 40 affairs (the operator command writes them: 91 of the 105 converter records carry pairs); next come the contents of the document parts marked by page and image detection (page precision 4/8). With OPD: its hash decision for the relinks and the 30-row spot-check kit (`results/opd_spotcheck/`); with the eCH group: `QAItem` and the five gaps. Author rules are per layout (held out: pair recall 39/137); where none exists the candidate finder, which offers 87 of the 141 gold names there, is the next lever. *Limits of the answer, each with its next test:* three splits rest on one parliament each, addressed body (Bellinzona 2/4), recommendation (Vernier 0/3) and language (Fribourg 4/6, OPD's tag), and on the affairs the gold and the Qwen check label alike the first two become shared (5/7 verdicts hold; post hoc): more parliaments per field. Italian is thin (no Ticino in P8, robots.txt; P10's primary cell German and French; P4 one commune): an Italian cell in P6's design. 6 fields are shared by definition only: their meaning test. **Reproducibility.** `make run` (Docker, CPU) recomputes the census and the Bern join, re-derives the P1 rules arm from the source PDFs, replays P1–P8 and P10 (all locks verified), regenerates the summary-number files byte for byte and runs the claims check, which binds every number from 10 up and every scope word in this report to its claim and checks this PDF against its source. **Declarations.** Claude Code agents (Anthropic) wrote most code, documentation and the gold labels. Public AI (may route outside Switzerland) served evaluation only. Report CC-BY-4.0.